

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually)

The classic Russian geroprotective protocol studied by Khavinson and colleagues combined Epithalamin and Thymalin, different bovine peptide extracts. The reported human outcomes were substantial but came from methodologically limited regional studies. This document evaluates a modern synthetic reinterpretation centered on Epitalon (AEDG) and Crystagen (EDP), with optional use of Pinealon or Vilon. The individual components have animal, cellular, and mechanistic support, while efficacy of the combined synthetic protocol remains speculative.

💉 Short-term safety of the individual peptides is plausible. Long-term safety of the complete combined protocol is speculative.

★ Epitalon is plausibly effective for geroprotective and longevity effects based on regionally concentrated animal evidence.

The efficacy of the complete synthetic peptide protocol is speculative.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 4-Oct-26. The most recent version of this document is available at <https://pdfs.researcher6076.com/pdfs/epitalon-longevity-protocol.pdf>



How Evidence Scope Was Applied in This Review

Evidence was evaluated here with an adjustment for source quality. Much of the Epitalon evidence comes from a single regional source, including small or methodologically limited studies that have not been independently replicated. When a claim depended primarily on this evidence, its evidence classification was reduced to reflect the additional uncertainty while still recognizing the evidence as informative.

Data support plausible short-term safety of the individual synthetic peptides. Frequent practitioner use without a consistent pattern of serious adverse effects is reassuring. Long-term safety of the combined synthetic protocol consisting of Epitalon, Crystagen or Vilon, and optional Pinealon is speculative. Telomere regulation, chromatin effects, and immune modulation create theoretical considerations that remain incompletely characterized.

Regionally concentrated studies support plausible geroprotective and longevity effects from Epitalon. Cellular research supports plausible telomere and cellular-repair effects where independent findings are available, while chromatin and broader gene-regulatory effects remain speculative. Efficacy of the complete synthetic protocol combining Epitalon with Crystagen or

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually)

Vilon and optional Pinealon is speculative. The reported regional human longevity evidence applies primarily to the extract-based Epithalamin and Thymalin regimen, not the synthetic protocol discussed here.

Extraordinary Claims with Important Caveats

Early Russian studies provide limited human evidence that protocols using Epithalamin from the pineal gland and Thymalin from the thymus likely improved immune balance, metabolic measures, and markers of biological aging.

Khavinson and Morozov reported outcomes from 266 adults over age 60 who were followed for 6–8 years at gerontology centers in St. Petersburg and Kyiv. Participants received Thymalin, Epithalamin, both extracts, or control treatment. Most treated participants received 10-day intramuscular courses during the first 2–3 years of observation. A separate group of 20 participants received both extracts annually for six years.

The investigators reported mortality rates 2.0–2.1 times lower with Thymalin, 1.6–1.8 times lower with Epithalamin, and 2.5 times lower with the combination than with control treatment. In the small group receiving both extracts annually for six years, mortality was reported as 4.1 times lower than in controls. The treated groups also showed improvements in reported immune, endocrine, cardiovascular, metabolic, and functional measures. Because these findings come from methodologically limited regional studies, they make benefit from the extract-based regimen likely rather than established.

Confidence in the exact magnitude of benefit is limited because the published account combines small cohorts and treatment schedules, provides less methodological detail than a modern trial report, and comes primarily from the source that developed the preparations. Some cohorts were described as stratified, randomized, and double-blind, but large, independent replication is lacking. The direction of benefit is likely, while the reported mortality reductions should be interpreted cautiously.

Crucially, **these trials used bovine extracts Epithalamin and Thymalin, not synthetic peptides. At best this evidence is suggestive for use of modern synthetic Epitalon and Crystagen.**

Replacing Epithalamin with Epitalon

Epitalon (AEDG) is a defined synthetic tetrapeptide developed from the Epithalamin research program, its name identifying its intended connection to the bovine pineal peptide complex. Its fixed sequence permits reproducible manufacture and direct mechanistic study. **Epitalon is a speculative partial analogue of Epithalamin but is not established as a clinically equivalent replacement for the extract.**

IBG Claims for Epitalon

The developers of Epitalon claim

- **Telomere modulation:** In vitro research, including an independent 2025 study, shows that Epitalon can increase telomere length in normal human epithelial and fibroblast cells through increased hTERT expression and telomerase activity. Telomere lengthening also occurred in breast-cancer cell lines, primarily through alternative lengthening of telomeres. These findings support a plausible effect on cellular telomere maintenance, while the organism-level longevity and cancer consequences remain uncertain.
- **Pineal and circadian regulation:** Epitalon plausibly supports pineal and circadian regulation based on regionally concentrated animal evidence showing effects on pineal secretion, melatonin-related rhythms, and age-associated neuroendocrine regulation. Improved human sleep or circadian function is speculative.
- **Antioxidant and mitochondrial protection:** Epitalon plausibly reduces oxidative stress based on cellular findings that include independent research. Broader mitochondrial protection remains speculative.
- **Chromatin and gene regulation:** Epitalon may speculatively influence chromatin organization and gene-expression programs involved in cellular maintenance and stress responses based on regionally concentrated cultured-cell studies.
- **Safety:** Limited, regionally concentrated human use and animal evidence support plausible short-term tolerability. Practitioner experience without a consistent pattern of serious adverse effects is reassuring, but long-term safety remains speculative.

Selecting a Synthetic Thymic Analogue

Thymalin supplied the thymic and immune-regulatory component of the original extract-based longevity protocol. It is a heterogeneous bovine thymic extract containing multiple biologically active short peptides, so no single synthetic peptide can be assumed to reproduce its complete effect.

Vilon (KE) and Crystagen (EDP) are defined short peptides associated with the Thymalin research program. Each may reproduce selected aspects of thymic or immune regulation, but neither is an established one-for-one replacement for Thymalin. Although named similarly, Thymulin is an unrelated, separate, naturally occurring zinc-dependent thymic hormone with distinct endocrine and immune-regulatory biology.

Candidate Evidence Profiles

Crystagen (EDP): Crystagen plausibly influences B-cell activity and splenic immune regulation based on a regionally concentrated animal study. Broader human immune and geroprotective applications are speculative. Its use as a complete replacement for Thymalin remains speculative.

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually)

Vilon (KE): Vilon plausibly has immune-regulatory and geroprotective effects based on regionally concentrated animal evidence. Effects on T-helper activity, chromatin organization, and broader immune signaling remain speculative. Human applications are speculative.

Thymulin: Zn-thymulin is the biologically active zinc-bound form of the naturally occurring thymic hormone. It participates in thymocyte maturation and immune regulation and is biologically distinct from the Khavinson short-peptide bioregulators. **Thymulin was not developed or classified as a Khavinson peptide bioregulator and has not been studied as a short-course thymic repair peptide.** It may be considered as a separate immune-regulatory adjunct rather than a synthetic replacement for Thymalin.

Selection among these peptides is theoretical and context dependent. No comparative study establishes one as the closest functional replacement for Thymalin.

<i>Attribute</i>	<i>Crystagen</i>	<i>Vilon</i>	<i>Thymulin</i>
Identity	Synthetic tripeptide Glu-Asp-Pro (EDP), described in the Thymalin peptide-research program	Synthetic dipeptide Lys-Glu (KE)	Natural zinc-dependent nonapeptide produced by thymic epithelial cells
Evidence-supported role	Plausibly influences B-cell and splenic immune regulation based on a regionally concentrated animal study	Plausibly influences T-helper activity, immune regulation, and geroprotective outcomes based on regionally concentrated animal evidence	Active Zn-thymulin participates in thymocyte maturation and T-cell regulation
Evidence scope	Human immune and geroprotective applications are speculative; disease-specific efficacy is speculative	Human immune and geroprotective applications are speculative; disease-specific efficacy is speculative	Its physiological role is established; benefits from administered zinc-free Thymulin remain plausible
Place in this protocol	Speculative candidate synthetic thymic component	Speculative alternative synthetic thymic component	Separate adjunct, not an interchangeable bioregulator or short-course repair peptide

Thymalin / Thymulin Confusion

The similar names “Thymalin” and “Thymulin,” together with vendor listings that combine or interchange the names, may contribute to substitution between biologically different products. Thymalin is a bovine thymic peptide extract and is unavailable through the gray or for-research-

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually)

use-only (FRU) peptide market; products offered in that market under similar names are Thymulin. Thymulin's thymic-endocrine and immune-regulatory effects do not make it a synthetic analogue of Thymalin or a short-course thymic repair peptide.

A Modern Synthetic Protocol

A modern synthetic reinterpretation may combine Epitalon with Crystagen or Vilon and optional Pinealon to address complementary pineal, thymic, immune, and neuro-metabolic pathways. The combined protocol and any additive effects remain speculative.

The 10- or 20-day course format is a speculative adaptation of the extract-based Russian regimen rather than an established schedule for the synthetic peptides. Either Crystagen or Vilon may be used as the thymic component, but comparative selection is speculative.

Pinealon (EDR) plausibly supports neuronal resilience and cognitive function based on regionally concentrated animal evidence showing improved spatial learning or retention and greater resistance to oxidative injury in specific rat models. Effects on reactive-oxygen accumulation, necrotic cell death, ERK signaling, cell-cycle activity, and NMDA-receptor gene expression remain speculative. Human cognitive and stress-resilience benefits are speculative. Including Pinealon after Epitalon and a thymic peptide is a speculative extension of the multi-axis framework.

Biological Mechanism

Epitalon: Epitalon plausibly supports pineal, circadian, and neuroendocrine regulation based on regionally concentrated animal evidence. Independent cellular studies support plausible telomerase, telomere-maintenance, oxidative-stress, and cellular-repair effects. Broader chromatin, mitochondrial, and gene-regulatory effects remain speculative.

Crystagen: Crystagen plausibly influences B-cell activity and splenic immune regulation based on a regionally concentrated animal study. Broader effects on thymic function, immune-cell turnover, and cytokine signaling remain speculative.

Vilon: Vilon plausibly influences T-helper activity and tumor outcomes based on regionally concentrated animal evidence. Effects on chromatin organization and broader immune signaling remain speculative.

Pinealon: Pinealon plausibly supports neuronal resilience and cognitive function based on regionally concentrated animal evidence. Effects on oxidative balance, necrotic cell death, ERK signaling, cell-cycle activity, and NMDA-receptor gene expression remain speculative.

Combining these peptides to produce complementary pineal, immune, and neuro-metabolic effects is speculative.

Dosage & Patterns

Human extract studies inform Epithalamin and Thymalin regimens but do not establish doses for synthetic Epitalon, Crystagen, Vilon, Pinealon, or their combination. Larger subcutaneous doses discussed by Western clinics and community users are speculative practice patterns. Community experience without a consistent pattern of serious adverse effects is reassuring but is not evidence of dose safety or equivalence.

<i>Component</i>	<i>Evidence-linked or regional pattern</i>	<i>Speculative community adaptation</i>	<i>Protocol role</i>
Epitalon	No established synthetic human regimen.	Synthetic Epitalon 5–10 mg daily for 10–20 days, often once or twice annually	Primary pineal component
Crystagen	Reports describe PO 0.5–1 mg daily for 10 days.	0.5–2 mg SC daily for a short course	Candidate thymic component
Vilon	Reports describe PO 100–200 mcg daily for 3–10 days.	1–5 mg SC daily for 10 days	Alternative thymic component
Pinealon	Reports describe PO 300 mcg daily for 10 days.	1–10 mg SC daily for 10 days	Optional CNS-focused component

Speculative combined-protocol patterns

Short daily courses: commonly described as lasting 10 days. Three-day and 20-day variations are also reported.

Sequential pattern: uses Epitalon first, followed by Crystagen or Vilon, with optional Pinealon afterward.

Overlapping optional: A partially overlapping pattern is discussed which begins the next peptide midway through the preceding peptide's course.

Repeating the sequence: once or twice annually is extrapolated from extract-based and Epitalon traditions, but without evidence.

Administration: Published Russian administration for Epitalon was commonly intramuscular, Crystagen and Vilon administration oral. Community administration is usually subcutaneous and equivalent exposure between these routes is uncertain.

Dose Controversy (Epitalon)

Human geroprotective studies commonly administered the Epithalamin extract at 10 mg intramuscularly for 10-day courses. Epitalon is a defined synthetic tetrapeptide, and its cellular and animal studies used varied, model-specific doses and routes.

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually)

Some discussions create the impression that the extract is “weaker” per mg, **but mass-based comparisons are misleading**. Because **extract doses are expressed as crude mass**, not quantified synthetic peptide, **a larger extract dose may still contain less of the specific active peptide than a small synthetic peptide dose**.

No validated conversion exists between the mass of the heterogeneous Epithalamin extract and the mass of pure synthetic Epitalon. The extract contains multiple peptides, and its absorption, proteolysis, and tissue exposure may differ from those of the isolated tetrapeptide.

Current Khavinson research publications discussing longevity specifically discuss Epitalon at 5 - 10 mg / day × 10 days as the most common dose, in a cycle that repeats once or twice annually, the same dose used for Epithalamin in their original research.

Doses widely described by Western clinics, practitioners, and community users for synthetic Epitalon are 5–10 mg daily for 10–20 days, repeated once or twice annually.

Both the Khavinson and practitioner described doses are speculative adaptation of the extract-based regimen rather than a dose established in a controlled human Epitalon trial.

Dose Variations (Crystagen)

Published evidence does not establish an injected Crystagen dose. Regional oral reports do not establish dose equivalence or provide a basis for subcutaneous dose selection.

Subcutaneous doses of 0.5–1 mg daily for 10 days are commonly discussed by Western practitioners and community users, and 2 mg daily is also reported. These are speculative practice patterns without comparative safety, pharmacokinetic, or efficacy data.

Dose Variations (Pinealon)

Regional Pinealon literature commonly describes oral doses of approximately 0.3 mg daily for 10 days. Transparent controlled human dose-ranging, pharmacokinetic, and injected-safety data are limited.

Western clinics and community users report doses ranging from 1–10 mg SC daily. The absence of a consistent pattern of serious adverse-event reports is reassuring but is not evidence that this doses is safe or equivalent. All SC dose patterns remain speculative.

Dose Variations (Vilon)

Regional Vilon literature and practice commonly describe an oral dose of approximately 100–200 micrograms daily for 3–10 days.

Western clinics and community users report doses of approximately 1–5 mg SC daily, commonly for 10 days. These speculative practice patterns and community tolerability reports do not establish their safety, efficacy, or equivalence to the regional regimens.

Low Dose Epitalon for Sleep

Epitalon's effects on pineal secretion and circadian regulation in regionally concentrated animal studies suggest a speculative mechanism for improved sleep timing or continuity. **Human evidence of improved melatonin rhythms primarily involve the Epithalamin extract and short treatment courses rather than nightly synthetic Epitalon.**

Western practitioners and community users report approximately 100–300 micrograms of Epitalon before sleep, administered subcutaneously or intranasally, either continuously or in cycles. This is a speculative community practice rather than a clinically studied sleep regimen.

Some users report that the perceived effect diminishes with nightly use and respond by cycling. Combining low-dose Epitalon with DSIP is also reported. Adaptation, cycling requirements, and additive sleep effects with DSIP remain speculative.

Benefits

The benefits below draw on different evidence bases. Human geroprotective and mortality findings primarily involve the Epithalamin and Thymalin extracts and their applicability to Epitalon are speculative at best. Synthetic Epitalon, Crystagen, Vilon, and Pinealon are supported mainly by animal and cellular research.

1. Cellular Aging, Telomeres & Genomic Stability

- **Telomere maintenance:** Epitalon plausibly increases telomerase activity and supports telomere maintenance in normal human cells, which may help preserve cellular replicative capacity.
- **Antioxidant and cellular protection:** Epitalon plausibly reduces oxidative stress based on cellular and animal findings that include independent cellular research. Broader mitochondrial and genomic protection remains speculative.
- **Chromatin and gene regulation:** Effects of Epitalon on chromatin organization and gene-expression programs involved in cellular maintenance, repair, and stress responses remain speculative.
- **Antimutagenic and antitumor effects:** Epitalon plausibly has antimutagenic and tumor-suppressive effects based on regionally concentrated animal studies. Its effects on human cancer risk are uncertain and may depend on tissue, tumor biology, and the telomere-maintenance mechanism involved.

2. Circadian, Pineal & Endocrine Regulation

- **Circadian and melatonin regulation:** Epitalon plausibly supports pineal secretion and circadian regulation based on regionally concentrated animal evidence. Improved human sleep timing, sleep continuity, and endocrine coordination are speculative.

3. Immune Function & Inflammatory Modulation (Crystagen-Dominant)

- **Crystagen immune regulation:** Crystagen plausibly influences B-cell activity and splenic immune regulation based on a regionally concentrated animal study. Broader human immune benefits are speculative.
- **Vilon immune regulation:** Vilon plausibly influences T-helper activity and immune regulation based on regionally concentrated animal evidence. Broader human immune benefits are speculative.

4. Longevity & Healthspan Signals

- **Geroprotective and longevity effects:** Epitalon plausibly has geroprotective and lifespan-supporting effects based on regionally concentrated animal evidence.
- **Combined synthetic protocol:** Complementary or additive effects from combining Epitalon with Crystagen or Vilon and optional Pinealon remain speculative.

5. Ocular & Retinal Support

- **Retinal cellular support:** Epitalon may plausibly support retinal-cell survival and repair through effects on cellular stress responses, cell migration, and wound closure observed in experimental retinal models.

6. Neuroprotection, Cognitive Resilience, and Synaptic Support (Pinealon-dominant)

- **Neuronal resilience and protection:** Pinealon plausibly improves neuronal resistance to oxidative injury based on regionally concentrated animal evidence.
- **Cellular signaling:** Effects of Pinealon on reactive-oxygen accumulation, necrotic cell death, ERK signaling, cell-cycle activity, and NMDA-receptor gene expression remain speculative.
- **Cognitive function:** Pinealon plausibly supports spatial learning or retention based on regionally concentrated animal evidence. Human cognitive benefits are speculative.

Potential Applications

- **Age-related immune decline:** Crystagen plausibly influences B-cell and splenic immune regulation based on a regionally concentrated aging-animal study. Its use for human age-related immune decline is speculative.
- **Broader immune dysfunction:** Support for impaired or dysregulated human immune function from Crystagen or Vilon is speculative.

Cautions

- **Organ transplantation or immunosuppressive therapy:** Immune-modulating peptides could plausibly interfere with intended immunosuppression.
- **Active malignancy:** The clinical implications of thymic immune modulation and peptide-related growth-regulatory effects are uncertain.

Side Effects

- **Community-reported effects:** Fatigue and headache are occasionally reported with use of these peptides, especially at higher doses.

Cancer Considerations

Epitalon: Epitalon plausibly exerts antitumor effects based on regionally concentrated animal studies reporting delayed tumor appearance and reduced tumor burden or metastasis. It also affects telomere-maintenance pathways, including telomerase activity in normal human cells and alternative lengthening of telomeres in breast-cancer cell lines. **Because of these mixed signals implications for an existing human malignancy are uncertain.**

Crystagen: Effects of Crystagen on broader thymic and immune regulation remain speculative. Any cancer-related benefit is speculative, and the **consequences of its immune-modulating effects during active malignancy are uncertain.**

Vilon: Vilon has produced differing cancer-related outcomes across regionally concentrated animal models, including reduced spontaneous neoplasms in CBA mice but increased mammary-tumor incidence and shorter tumor latency in HER-2/neu transgenic mice. Its effects may plausibly depend on tumor biology and disease context; **Because of these mixed signals the implications for human malignancy are uncertain.**

Clean Window

Some practitioners use a “clean window” by minimizing concurrent experimental peptides or major protocol changes before, during, and shortly after this protocol. This practice is speculative and is intended to reduce confounding, improve attribution of effects, and avoid overlapping biological signals. Timing decisions are individualized.

Conclusions

Epitalon plausibly has geroprotective and lifespan-supporting effects based on regionally concentrated animal evidence. Combining Epitalon with Crystagen or Vilon and optional Pinealon is a speculative modern reconstruction of the broader pineal and thymic approach. Short-term safety of the individual peptides is plausible, and long-term safety of the combined protocol is speculative. Practitioner experience is reassuring but is not scientific evidence.

References

1. Al-Dulaimi S, Thomas R, Matta S, Roberts T. Epitalon increases telomere length in human cell lines through telomerase upregulation or ALT activity. *Biogerontology*. 2025;26(5):178. doi:10.1007/s10522-025-10315-x. PMID: 40908429.
2. Gatta M, Dovizio M, Milillo C, et al. The antioxidant tetrapeptide Epitalon enhances delayed wound healing in an in vitro model of diabetic retinopathy. *Stem Cell Rev Rep*. 2025;21:1822-1834. doi:10.1007/s12015-025-10911-x.
3. Khavinson V, Goncharova N, Lapin B. Synthetic tetrapeptide Epitalon restores disturbed neuroendocrine regulation in senescent monkeys. *Neuro Endocrinol Lett*. 2001;22(4):251-254. PMID: 11524632.

4. Anisimov VN, Khavinson VK, Popovich IG, et al. Effect of Epitalon on biomarkers of aging, life span and spontaneous tumor incidence in female Swiss-derived SHR mice. *Biogerontology*. 2003;4(4):193-202. doi:10.1023/A:1025114230714. PMID: 14501183.
5. Anisimov VN, Khavinson VK, Provinciali M, et al. Inhibitory effect of the peptide Epitalon on the development of spontaneous mammary tumors in HER-2/neu transgenic mice. *Int J Cancer*. 2002;101(1):7-10. doi:10.1002/ijc.10570. PMID: 12209581.
6. Kossoy G, Anisimov VN, Ben-Hur H, Kossoy N, Zusman I. Effect of the synthetic pineal peptide Epitalon on spontaneous carcinogenesis in female C3H/He mice. *In Vivo*. 2006;20(2):253-257. PMID: 16634527.
7. Anisimov VN, Popovich IG, Zabezhinskii MA, et al. Effect of Epitalon and melatonin on life span and spontaneous carcinogenesis in senescence-accelerated mice. *Vopr Onkol*. 2005;51(1):93-98. PMID: 15909815.
8. Khavinson VK, Morozov VG. Geroprotective effect of Thymalin and Epithalamin. *Adv Gerontol*. 2002;10:74-84. PMID: 12577695.
9. Khavinson VK, Lezhava TA, Malinin VV. Effects of short peptides on lymphocyte chromatin in senile subjects. *Bull Exp Biol Med*. 2004;137(1):78-81. doi:10.1023/B:BEBM.0000024393.40560.05. PMID: 15085253.
10. Khavinson VK, Anisimov VN, Zavarzina NY, et al. Effect of Vilon on biological age and lifespan in mice. *Bull Exp Biol Med*. 2000;130(7):687-690. doi:10.1007/BF02682106. PMID: 11140587.
11. Khavinson VK, Anisimov VN. A synthetic dipeptide Vilon (L-Lys-L-Glu) inhibits growth of spontaneous tumors and increases life span of mice. *Dokl Biol Sci*. 2000;372:261-263. PMID: 10944717.
12. Chervyakova NA, Linkova NS, Chalisova NI, Koncevaya EA, Trofimova SV, Khavinson VK. Molecular aspects of immunoprotective activity of peptides in spleen during the ageing process. *Adv Gerontol*. 2014;27(1):224-228. PMID: 28976144.
13. Khavinson V, Ribakova Y, Kulebiakin K, et al. Pinealon increases cell viability by suppression of free-radical levels and activation of proliferative processes. *Rejuvenation Res*. 2011;14(5):535-541. doi:10.1089/rej.2011.1172. PMID: 21978084.
14. Arutjunyan A, Kozina L, Stvolinskiy S, Bulygina Y, Mashkina A, Khavinson V. Pinealon protects rat offspring from prenatal hyperhomocysteinemia. *Int J Clin Exp Med*. 2012;5(2):179-185. PMID: 22567179.
15. Karantysh GV, Fomenko MP, Menzheritskii AM, Prokof'ev VN, Ryzhak GA, Butenko EV. Effect of Pinealon on learning and expression of NMDA receptor subunit genes in the hippocampus of rats with experimental diabetes. *Neurochem J*. 2020;14:314-320. doi:10.1134/S181971242003006X.
16. Dardenne M, Pleau JM. Interactions between zinc and Thymulin. *Met Based Drugs*. 1994;1(2-3):233-239. doi:10.1155/MBD.1994.233. PMID: 18476235.

Epitalon (AEDG) + Crystagen (EDP-3) + Pinealon (EDR) Available for Research Use Only
(*Epitalon 5 - 10 mg × 10 days, Crystagen 0.5 – 1 mg / day × 10 days, Pinealon ≤ 0.5 mg / day × 10 days, 1 – 2 × annually*)

17. Khavinson V, Linkova N, Kozhevnikova E, Dyatlova A, Petukhov M. Transport of biologically active ultrashort peptides using POT and LAT carriers. *Int J Mol Sci.* 2022;23(14):7733. doi:10.3390/ijms23147733. PMID: 35887081.